



mem9 shared memory · on TiDB Cloud

v3

SOLUTION BRIEF · TECHNICAL EVALUATION

Agentic Coding - Shared Memory Layer

One shared memory for a team of agents.

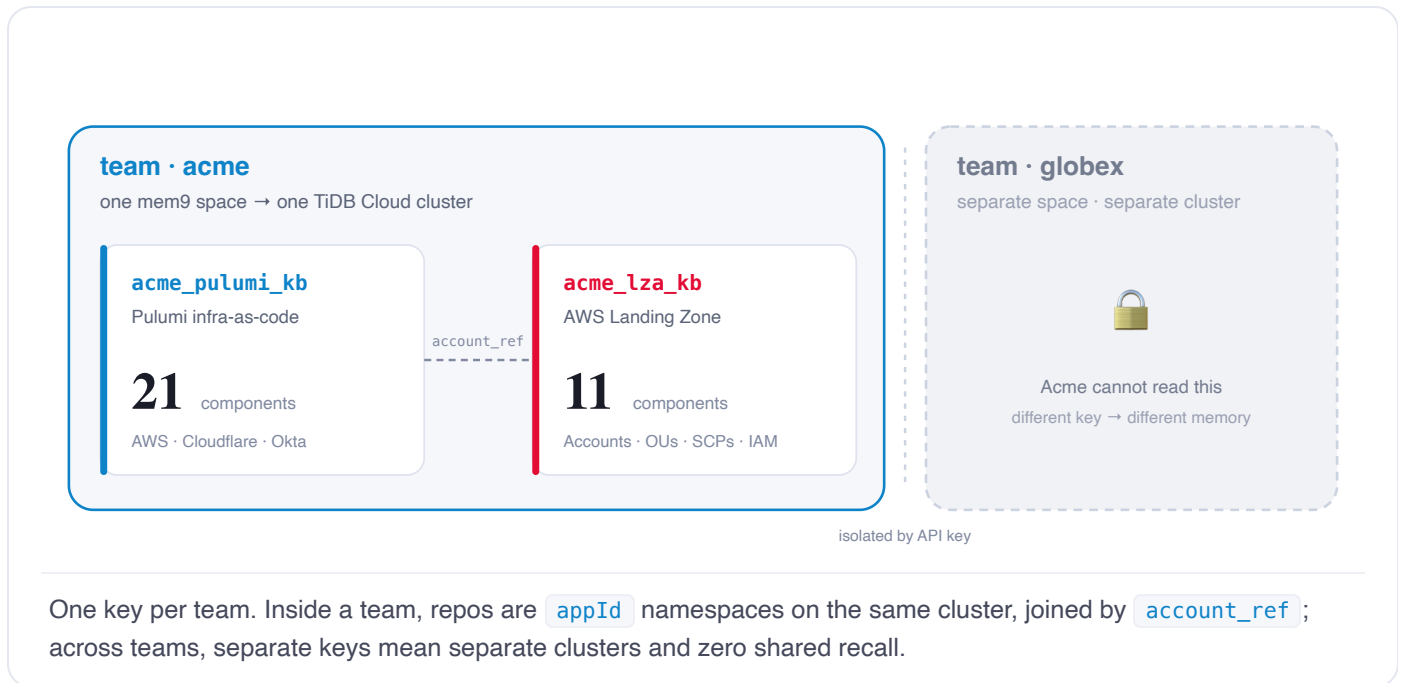
Bottom line. mem9 gives a team of AI coding agents one shared, queryable memory on TiDB Cloud. Agents read before they build and write back when done - so none starts cold or duplicates work. It is isolated per team, an agent can work across repos in a single session, and it answers the one question a vector store cannot: the transitive blast radius of a change. Every screenshot in this brief is a live read or write against TiDB Cloud.

THE PROBLEM	HOW MEM9 SOLVES IT	SHOWN IN
Each team's knowledge must stay private - no team reading another's.	One TiDB Cloud cluster per team (a mem9 space). A team's key reaches only its own memory; isolation is enforced by the platform, not application code.	§ The model
One integration should serve many repositories, not one wiring per repo.	Within a team, repos are <code>appId</code> namespaces on one cluster and the agent routes to the right one . Across teams, separate keys mean separate clusters.	§ The model
A single task often spans repos - create a resource in one, the dependent resource in another.	One agent session writes to each repo's namespace and links the records by <code>account_ref</code> - one integration, two repos, no manual hand-off.	§ One task, two repos
Some teams require a self-hosted deployment.	Supported. Vector and graph run self-hosted; full-text search is TiDB Cloud only today.	§ Deployment

Demonstrated live with three different coding agents – Claude Code, Minimax, and Cursor – against a single TiDB Cloud cluster. The tools are interchangeable; the shared memory is the point.

Team = cluster. Repo = namespace.

mem9 maps onto how a team already works. A **team** is a mem9 **space** - one key, one isolated TiDB Cloud cluster. A **repo** is an `appId` namespace inside it, so repos share a cluster but keep separate recall scopes. Another team is another cluster, reachable only with its own key.



Each team's space sits on its own managed TiDB Cloud cluster. Agents never handle raw database credentials – the mem9 API key is the only secret.

One task, two repos.

A single task often spans repos - create a new AWS account in the LZA repo *and* the S3 bucket for it in the Pulumi repo. One agent, given one plain-language request, writes the account to `acme_lza_kb` and the bucket to `acme_pulumi_kb` - two separate namespaces - and links them with `account_ref`. No second session, no manual hand-off.

mems

Infra Knowledge Base

acme · pulumi + lza

TIDB Cloud

Reset

Live

Cross-repo

Dependencies

Search

Config

One task. Two repos. One memory.

A single agent session provisions a new AWS account in the **LZA** repo and an S3 bucket in the **Pulumi** repo deployed into it - reading and writing across both `appId` namespaces in one MCP session, joined by `account_ref`.

Run cross-repo task

dev types: "Create a new data-platform AWS account in LZA and an exports bucket for it in Pulumi."

cross-repo task complete

ACTIVITY · BOTH REPOS

claude-code

WRITE

pulumi

Created acme-prod-data-platform-exports, an S3 bucket deployed into the new data-platform account.

claude-code

WRITE

lza

Created acme-lza-account-data-platform, a new AWS account in the workloads OU.

ACCOUNT_REF JOIN · LZA ACCOUNT ↔ PULUMI RESOURCES

lza

account-data-platform

new

pulumi

data-platform-exports

account_ref=data-platform

lza

account-prod

account_ref=prod

pulumi

analytics-db

db

data-exports

s3

static-assets

s3

assets-dns

Cloudflare

admin-dns

Cloudflare

admin-sso

Okta

+6 more

lza

account-sandbox

account_ref=sandbox

pulumi

analytics-db

db

data-exports

s3

DASHBOARD · CROSS-REPO Left: one activity feed shows the same agent writing to **both** repos. Right: the `account_ref` join - the new **data-platform** account and its Pulumi exports bucket, created together and linked.

Why it matters. The agent doesn't need to be told which repo owns what. It routes each write to the right namespace and records the cross-repo link, so the next session - in either repo - already knows the account and its bucket belong together.

Three tools, one memory, warm hand-offs.

A common scenario: production is complete, staging has drifted by four components. Three different agents bring it back to parity - each reading the shared memory before it builds and writing back when done. The developer only states the goal; the standing instruction does the rest.

1 • Claude Code

```
dev types "Bring staging up to parity with production - start with the static-assets bucket and its CDN."
```

Queries the KB, finds the gap, composes the libraries, writes both back. **Gap 4 → 2.**

2 • Minimax

```
dev types "Keep bringing staging to parity with prod - it still needs the admin-portal DNS."
```

Opens cold. Reads the shared session log, sees what was just built, continues with admin-DNS. **Gap 2 → 1.**

3 • Cursor

```
dev types "Finish bringing staging to parity - add the admin SSO app."
```

Traces what prod's admin-SSO depends on, confirms it redirects to a DnsRecord, builds the SSO app. **Gap 1 → 0.**

The gap closing - real writes to mem9, each tool credited:

mem9

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acme · pulumi + 12a

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Three tools, one memory, live.

Every query and write from Claude Code, Minimax, and Cursor lands in the same mem9 space. Watch the staging gap close as each tool reads the shared log and writes back.

ACTIVITY

No activity yet. Launch a CLI and ask it to query the knowledge base.

KNOWLEDGE GRAPH

All Lib Prod Staging Org



STAGING PARITY

4 missing in staging

✓ analytics-db	RDS	in staging
✓ data-exports	S3	in staging
✗ static-assets	S3	missing
✗ assets-dns	Cloudflare	missing

START

Feed empty; staging-parity strip shows **4 missing**.

mem9

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acme · pulumi + lza

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Three tools, one memory, live.

Every query and write from Claude Code, Minimax, and Cursor lands in the same mem9 space. Watch the staging gap close as each tool reads the shared log and writes back.

ACTIVITY

claude-code

WRITE

Created acme-staging-assets-dns, a Cloudflare DNS fronting acme-staging-static-assets.

claude-code

WRITE

The acme-staging-static-assets resource is an S3 bucket composing the S3Bucket library.

KNOWLEDGE GRAPH

All

Lib

Prod

Staging

Org

Library

S3

RDS

Cloudflare

Okta

Account

2 missing in staging

✓ analytics-db	RDS	in staging
✓ data-exports	S3	in staging
✓ static-assets	S3	in staging
✓ assets-dns	Cloudflare	in staging

1 · CLAUDE CODE Reads the KB, composes **static-assets** and its CDN, writes both back. Gap 4 → 2.

mem9

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Three tools, one memory, live.

Every query and write from Claude Code, Minimax, and Cursor lands in the same mem9 space. Watch the staging gap close as each tool reads the shared log and writes back.

ACTIVITY

minimax

WRITE

Created acme-staging-admin-dns, a Cloudflare DNS matching the production admin pattern.

minimax

QUERY

The user is continuing staging work.

claude-code

WRITE

Created acme-staging-assets-dns, a Cloudflare DNS fronting acme-staging-static-assets.

claude-code

WRITE

The acme-staging-static-assets resource is an S3 bucket composing the S3Bucket library.

KNOWLEDGE GRAPH

All Lib Prod Staging Org

Library S3 RDS Cloudflare Okta Account

pulumi 12a

STAGING PARITY

1 missing in staging

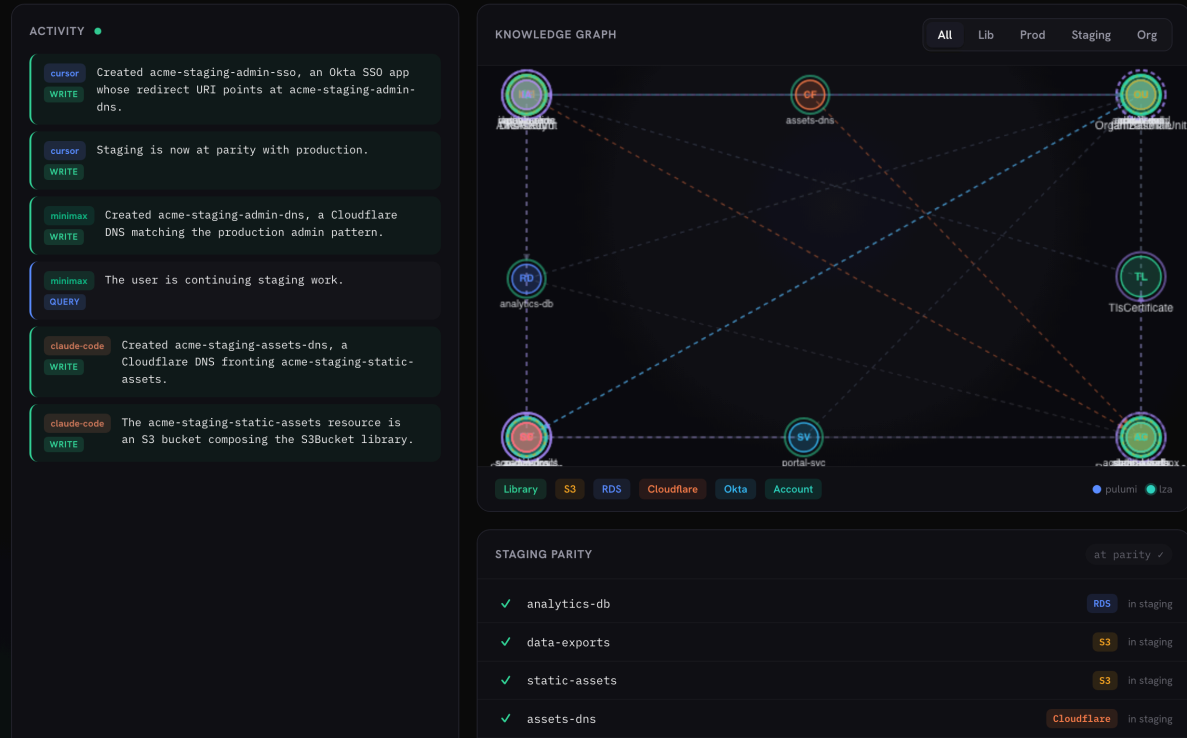
✓ analytics-db	RDS	in staging
✓ data-exports	S3	in staging
✓ static-assets	S3	in staging
✓ assets-dns	Cloudflare	in staging

2 · MINIMAX

Opens cold, reads the shared log first ("continuing staging work"), then builds **admin-DNS** - no re-briefing. Gap 2 → 1.

Three tools, one memory, live.

Every query and write from Claude Code, Minimax, and Cursor lands in the same mem9 space. Watch the staging gap close as each tool reads the shared log and writes back.

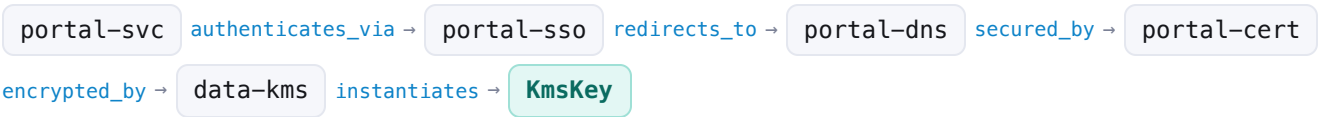


3 · CURSOR

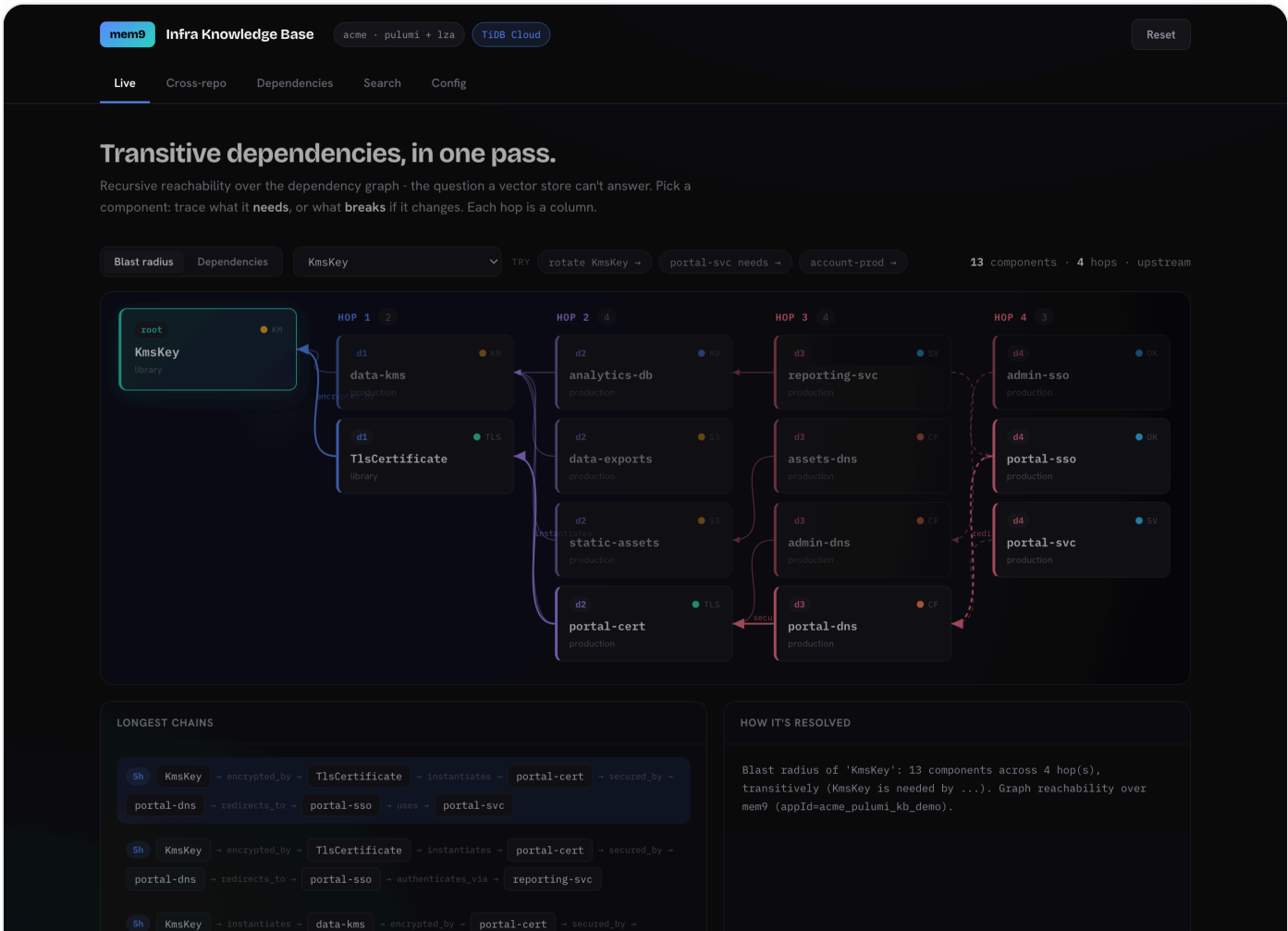
Traces what admin-SSO depends on, builds the **SSO app**; the last write reads "Staging is now at parity with production." Gap **1** → **0**.

The question similarity can't answer.

A vector store finds things that *look* related. It cannot compute a transitive closure - "what, exactly, breaks if I change this?" That is graph reachability. mem9 records every `depends_on` and composition edge; one recursive traversal walks the chain to any depth.



Run that in reverse from `KmsKey` - "what breaks if we rotate it?" - and the answer is **13 components across four depths**. No flat search surfaces this; only walking the graph does.



DASHBOARD · DEPENDENCIES "What breaks if we rotate **KmsKey**?" Each hop is a column; 13 components fan across four depths and funnel onto the anchored foundation. The Trace panel reads each chain as a sentence.

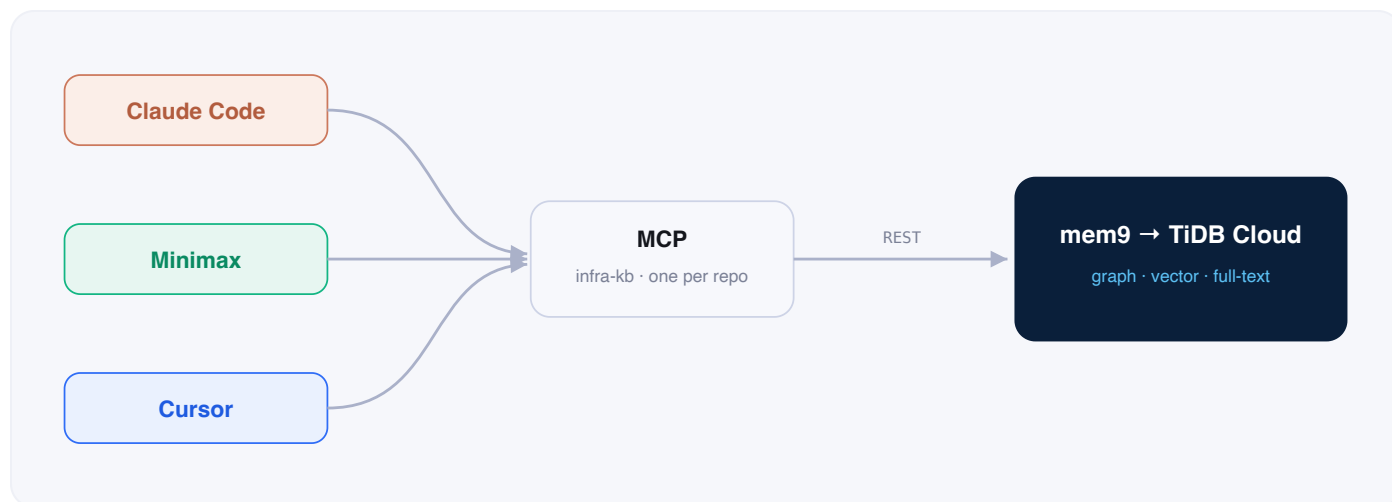
One engine, not four

APPROACH	GRAPH REACHABILITY	SEMANTIC RECALL	ATOMIC AGENT STATE
S3 / filesystem	no traversal	no	no transactions
Vector DB only	similarity ≠ reachability	yes	no
Four separate systems	partial, with ETL	yes	no cross-system atomicity
mem9 → TiDB Cloud	recursive traversal	vector + full-text	one transaction

Graph, vector embeddings, and full-text live in the same store - the same recall that answers "*what encrypts production data*" also computes the blast radius above. No separate vector DB, no ETL, consistent across concurrent writes from all three tools.

S3 is the filing cabinet for artifacts. mem9 is the brain agents query.

One store. One plug. One key.



MCP is the plug

Every tool loads the same Model Context Protocol config - one named server per repo. No custom glue per tool.

One store, every pattern

Relational state, the dependency graph, vector embeddings, and full-text all live in the same memory.

Just an API key

mem9 manages the TiDB Cloud cluster. Agents read and write through one key and never touch credentials.

The standing instruction

The developer types a normal request. The prescriptive part lives in each tool's rules file (`CLAUDE.md` / `AGENTS.md` / `.cursorrules`), so the agent queries first, composes, and writes back on every run with no per-task hand-holding.

mem9

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Behavior is configuration, not code.

The developer just types a normal request - "bring staging to parity." This standing instruction, loaded into every tool's rules file, makes the agent query first, compose, and write back on its own. No per-task hand-holding.

STANDING INSTRUCTION · CLAUDE.MD / AGENTS.MD / .CURSORRULES

Before you write or change any infrastructure - automatically, every task:

1. Query first. query_knowledge_base on the infra-kb MCP for the repo you're touching: what exists, how it connects, recent log.
2. Compose. Build from libs/ (S3Bucket, PostgresDatabase, DnsRecord, SsoApplication, KmsKey, ...). Never raw aws.*/cloudflare.*/okta.*.
3. Conventions. acme-<env>-<name> naming; required tags; DnsRecord proxied; SSO redirect -> DnsRecord; set account_ref (prod/sandbox).
4. Blast radius. Before changing a shared component, trace what depends on it.
5. Write back. After any change, call write_component to record it - or the next tool, and the next session, start cold.

The human never says these steps. They say what they want; you do this on your own.

MCP SERVERS · ONE ENTRY PER REPO

```
{
  "mcpServers": {
    "infra-kb-pulumi": { "command": "python", "args": ["-m", "src.mcp_server"] },
    "infra-kb-lza": { "command": "python", "args": ["-m", "src.mcp_server"] }
  }
}
```

query_knowledge_base · write_component -> hybrid recall + atomic writes

DASHBOARD · CONFIG

The five-step standing instruction every tool loads, and the one-entry-per-repo MCP wiring that makes it automatic.

Deployment

- **TiDB Cloud (shown here):** graph + vector + full-text, fully managed; nothing to host.
- **Self-hosted:** available - vector and graph run on-prem. Full-text search is TiDB Cloud only today, so this brief uses Cloud to show hybrid recall end to end.

mem9 · TiDB Cloud – shared memory for AI coding agents · v3. Live demo: Claude Code · Minimax · Cursor over team acme (repos pulumi + lza). Every figure is a live read/write against TiDB Cloud.